SG No. 204 T_{h}^{5} $Im\bar{3}$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$

Table 1: Wyckoff site: `2a`, site symmetry: `m-3.`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `6b`, site symmetry: `mmm..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,6,7,8,17,18,19,20]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,11,12,21,22,23,24]`

Table 3: Wyckoff site: `8c`, site symmetry: `.-3.`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,5,9,13,17,21]`
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,7,12,14,19,24]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[4,6,11,16,18,23]`

Table 4: Wyckoff site: `12d`, site symmetry: `mm2..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4,14,15]`
2	$[-x,\ 0,\ 0]$	`[2,3,13,16]`
3	$[0,\ x,\ 0]$	`[5,8,18,19]`
4	$[0,\ -x,\ 0]$	`[6,7,17,20]`
5	$[0,\ 0,\ x]$	`[9,12,22,23]`
6	$[0,\ 0,\ -x]$	`[10,11,21,24]`

Table 5: Wyckoff site: `12e`, site symmetry: `mm2..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,4,14,15]`
2	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2,3,13,16]`
3	$\left[\frac{1}{2},\ x,\ 0\right]$	`[5,8,18,19]`
4	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[6,7,17,20]`
5	$\left[0,\ \frac{1}{2},\ x\right]$	`[9,12,22,23]`
6	$\left[0,\ \frac{1}{2},\ -x\right]$	`[10,11,21,24]`

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,9]`
2	$[-x, -x, x]$	`[2,7,12]`
3	$[-x, x, -x]$	`[3,8,10]`
4	$[x, -x, -x]$	`[4,6,11]`
5	$[-x, -x, -x]$	`[13,17,21]`
6	$[x, x, -x]$	`[14,19,24]`
7	$[x, -x, x]$	`[15,20,22]`
8	$[-x, x, x]$	`[16,18,23]`

Table 7: Wyckoff site: `24g`, site symmetry: `m..`

No.	position	mapping
1	$[0, y, z]$	`[1,16]`
2	$[0, -y, z]$	`[2,15]`
3	$[0, y, -z]$	`[3,14]`
4	$[0, -y, -z]$	`[4,13]`
5	$[z, 0, y]$	`[5,20]`
6	$[z, 0, -y]$	`[6,19]`
7	$[-z, 0, y]$	`[7,18]`
8	$[-z, 0, -y]$	`[8,17]`
9	$[y, z, 0]$	`[9,24]`
10	$[-y, z, 0]$	`[10,23]`
11	$[y, -z, 0]$	`[11,22]`
12	$[-y, -z, 0]$	`[12,21]`

Table 8: Wyckoff site: `48h`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[2]`
3	$[-x, y, -z]$	`[3]`
4	$[x, -y, -z]$	`[4]`
5	$[z, x, y]$	`[5]`
6	$[z, -x, -y]$	`[6]`
7	$[-z, -x, y]$	`[7]`
8	$[-z, x, -y]$	`[8]`
9	$[y, z, x]$	`[9]`
10	$[-y, z, -x]$	`[10]`
11	$[y, -z, -x]$	`[11]`
12	$[-y, -z, x]$	`[12]`
13	$[-x, -y, -z]$	`[13]`
14	$[x, y, -z]$	`[14]`
15	$[x, -y, z]$	`[15]`
16	$[-x, y, z]$	`[16]`

continued ...

Table 8

No.	position	mapping
17	$[-z, -x, -y]$	[17]
18	$[-z, x, y]$	[18]
19	$[z, x, -y]$	[19]
20	$[z, -x, y]$	[20]
21	$[-y, -z, -x]$	[21]
22	$[y, -z, x]$	[22]
23	$[-y, z, x]$	[23]
24	$[y, z, -x]$	[24]